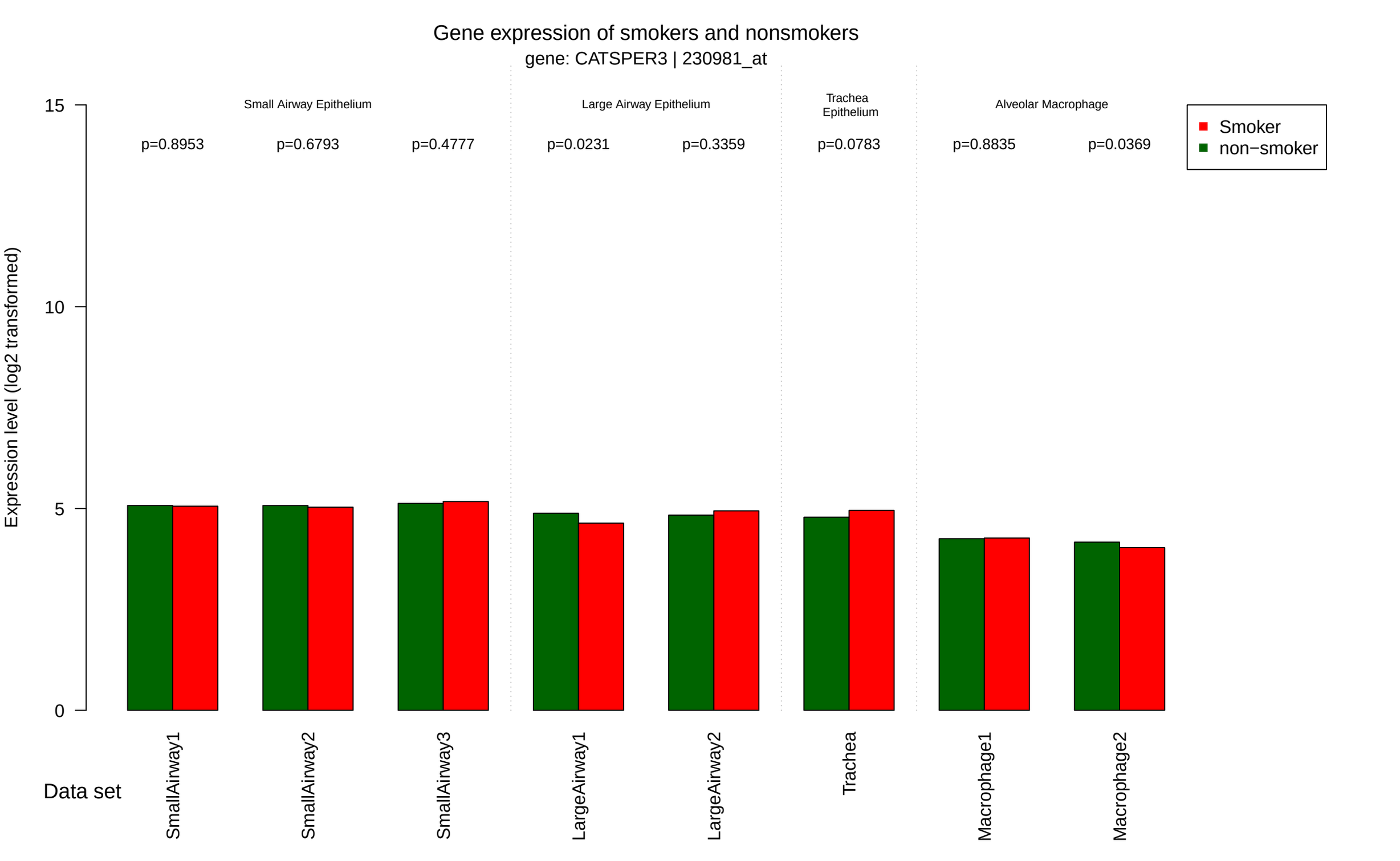
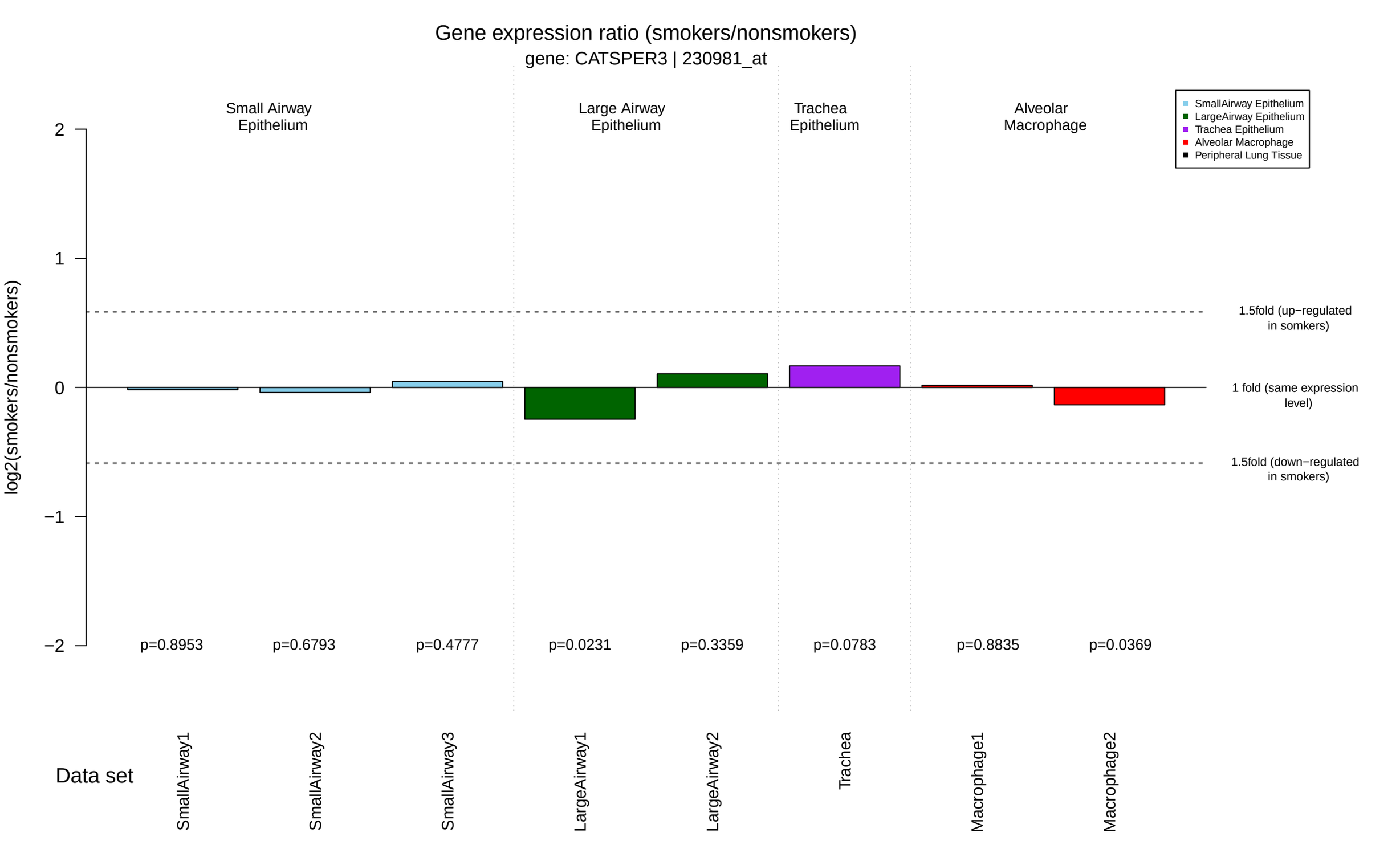
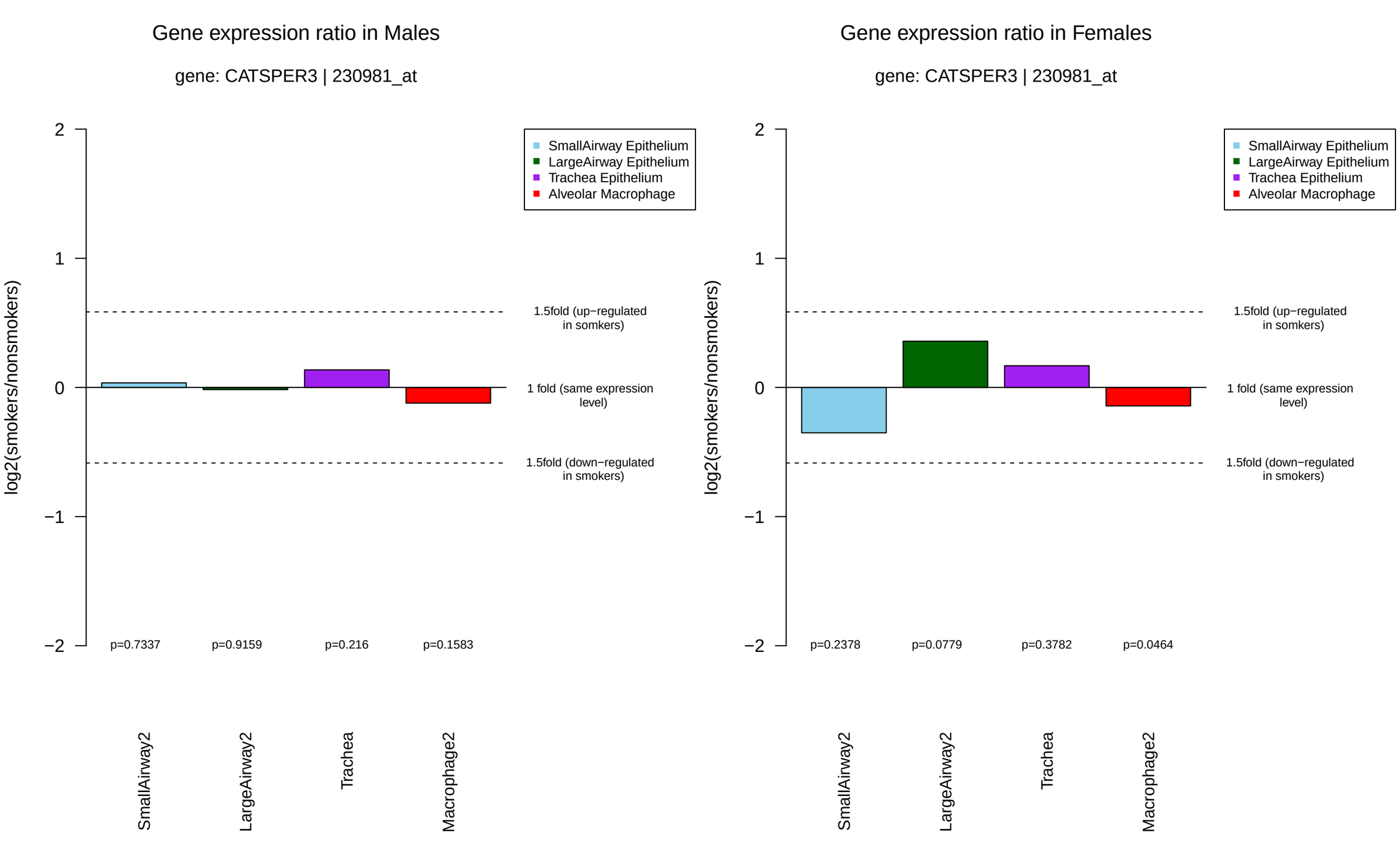
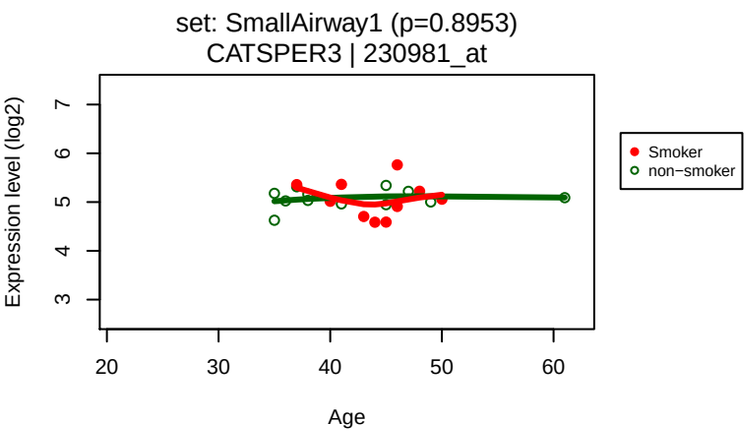
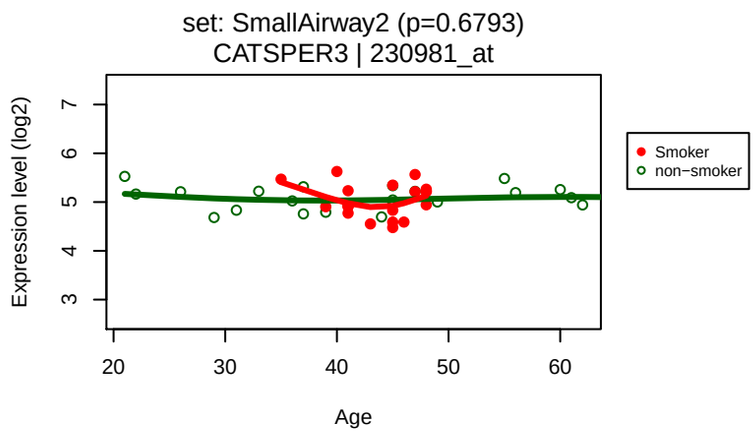
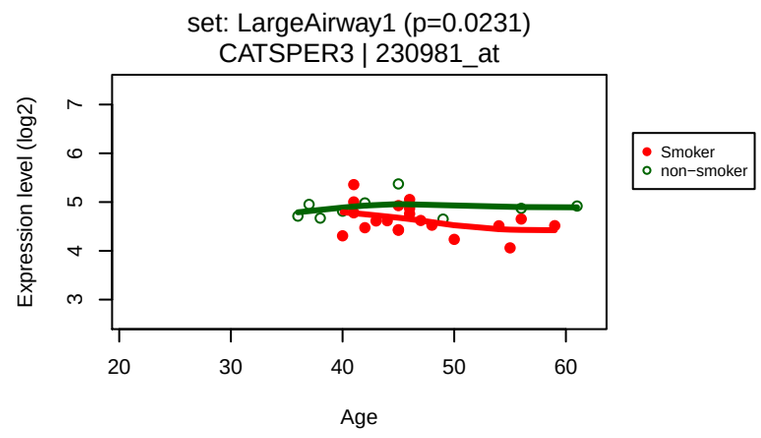

Gene expression of smokers and nonsmokers
gene: CATSPER3 | 230981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8953
p=0.6793
p=0.4777
p=0.0231
p=0.3359
p=0.0783
p=0.8835
p=0.0369
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

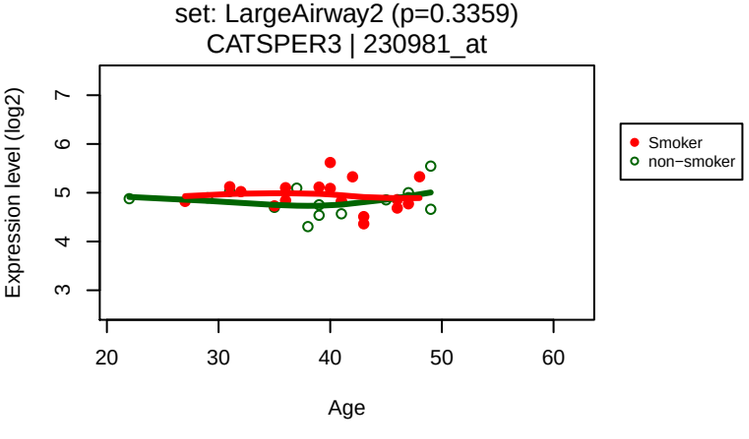

Gene expression ratio (smokers/nonsmokers)
gene: CATSPER3 | 230981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8953
p=0.6793
p=0.4777
p=0.0231
p=0.3359
p=0.0783
p=0.8835
p=0.0369
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

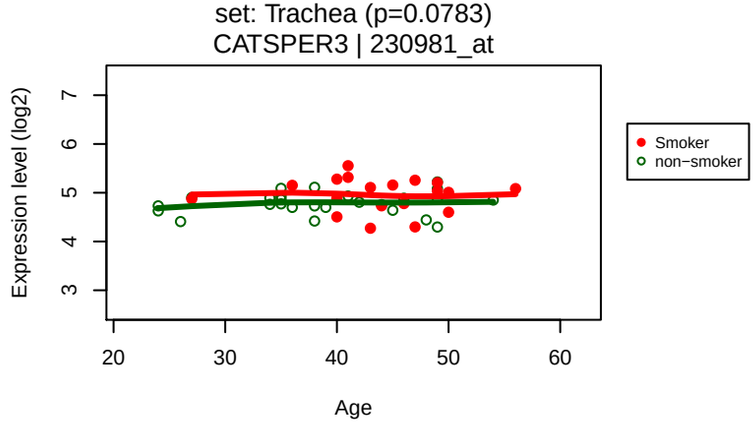

Gene expression ratio in Males
gene: CATSPER3 | 230981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7337
p=0.9159
p=0.216
p=0.1583
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CATSPER3 | 230981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2378
p=0.0779
p=0.3782
p=0.0464
SmallAirway2
LargeAirway2
Trachea
Macrophage2

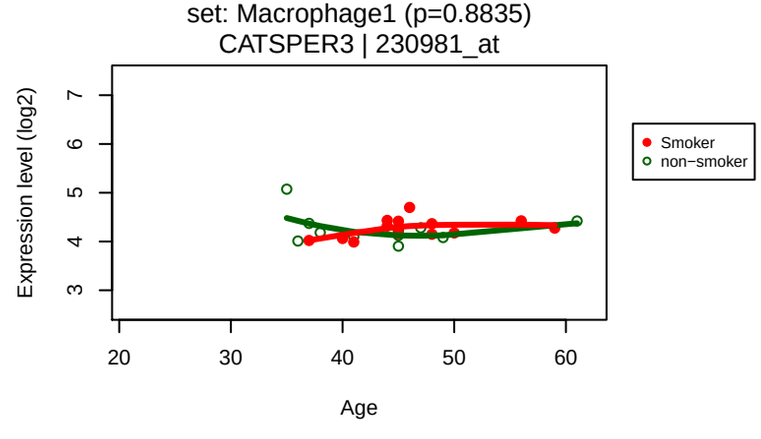

set: SmallAirway1 (p=0.8953)
CATSPER3 | 230981_at
Expression level (log2)
Age
Smoker
non−smoker

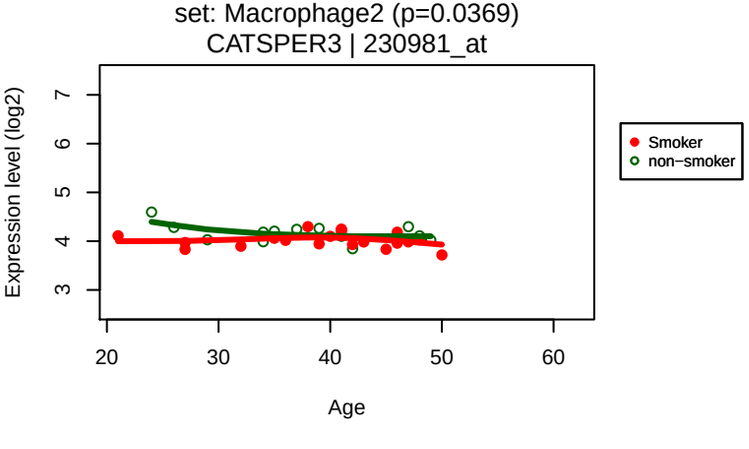

set: SmallAirway2 (p=0.6793)
CATSPER3 | 230981_at
Expression level (log2)
Age
Smoker
non−smoker

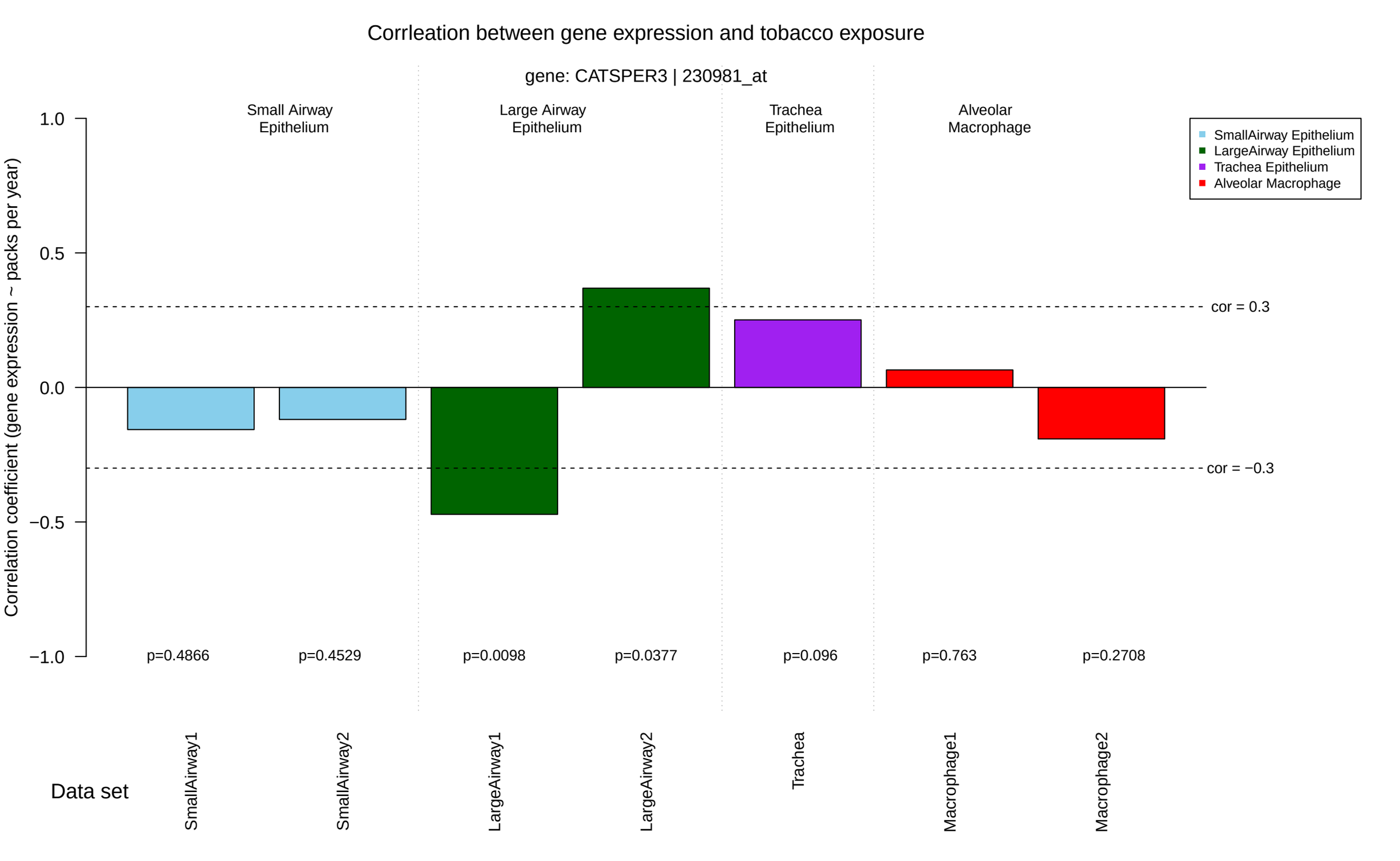

set: LargeAirway1 (p=0.0231)
CATSPER3 | 230981_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3359)
CATSPER3 | 230981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0783)
CATSPER3 | 230981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8835)
CATSPER3 | 230981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0369)
CATSPER3 | 230981_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CATSPER3 | 230981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4866
p=0.4529
p=0.0098
p=0.0377
p=0.096
p=0.763
p=0.2708
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2